

Predicting drug-target interactions using restricted Boltzmann machines

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ABSTRACT

Motivation: *In silico* prediction of drug-target interactions plays an important role toward identifying and developing new uses of existing or abandoned drugs. Network-based approaches have recently become a popular tool for discovering new drug-target interactions (DTIs). Unfortunately, most of these network-based approaches can only predict binary interactions between drugs and targets, and information about different types of interactions has not been well exploited for DTI prediction in previous studies. On the other hand, incorporating additional information about drug-target relationships or drug modes of action can improve prediction of DTIs. Furthermore, the predicted types of DTIs can broaden our understanding about the molecular basis of drug action.

Results: We propose a first machine learning approach to integrate multiple types of DTIs and predict unknown drug-target relationships or drug modes of action. We cast the new DTI prediction problem into a two-layer graphical model, called *restricted Boltzmann machine*, and apply a practical learning algorithm to train our model and make predictions. Tests on two public databases show that our *restricted Boltzmann machine* model can effectively capture the latent features of a DTI network and achieve excellent performance on predicting different types of DTIs, with the area under precision-recall curve up to 89.6. In addition, we demonstrate that integrating multiple types of DTIs can significantly outperform other predictions either by simply mixing multiple types of interactions without distinction or using only a single interaction type. Further tests show that our approach can infer a high fraction of novel DTIs that has been validated by known experiments in the literature or other databases. These results indicate that our approach can have highly practical relevance to DTI prediction and drug repositioning, and hence advance the drug discovery process.

Availability: Software and datasets are available on request.

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Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 INTRODUCTION

Drug development currently remains an expensive and time-consuming process with extremely low success rate: it typically takes 10–15 years and \$800 million–1 billion to bring a new drug to market (Dimasi, 2001). In recent decades, the rate of the number of new drugs approved by the US Food and Drug Administration versus the amount of money invested in pharmaceutical research and development has significantly declined (Booth and Zimmel, 2004). This productivity problem has

urged drug developers to seek new uses for existing or abandoned drugs (Booth and Zimmel, 2004). Such a new strategy is also called *drug repositioning* or *drug repurposing*. A strong support for the possibility of drug repositioning is the increasingly accepted concept of ‘polypharmacology’, i.e. individual drugs can interact with multiple targets rather than a single target (MacDonald *et al.*, 2006; Xie *et al.*, 2012). For example, serotonin and serotonergic drugs can interact with both 5-HT₁ G protein-coupled receptors and 5HT_{3A} ion channel proteins, even though these two target proteins are not related in sequence or structure (Cheng *et al.*, 2012; Keiser *et al.*, 2007; Kroeze *et al.*, 2002; Roth *et al.*, 2004). This polypharmacological property of a drug enables us to identify more than one target that it can act on and hence develop its new uses.

In silico prediction of interactions between drugs and target proteins provides an important tool for drug repositioning, as it can significantly reduce wet-laboratory work and lower the cost of the experimental determination of new drug-target interactions (DTIs). Various methods have been proposed for *in silico* DTI prediction. When 3D structures are available, molecular docking is commonly used to virtually screen a large number of compounds against a target protein (Cheng *et al.*, 2007; Donald, 2011; Morris *et al.*, 2009). When 3D structures of molecules are absent, a number of different approaches have been developed to address the *in silico* DTI prediction problem. Most of these structure-free methods can be grouped into two classes, namely, *ligand-based* and *network-based* approaches. A representative ligand-based approach is the *similarity ensemble approach* (Keiser *et al.*, 2007, 2009), which predicts new DTIs using 2D structure similarity of ligands. Although ligand-based approaches are able to discover a number of DTIs that have been validated experimentally, they have difficulty in identifying drugs with novel scaffolds that differ from those of reference compounds (Yabuuchi *et al.*, 2011).

Numerous network-based approaches have been proposed to exploit latent features of DTI profiles and have recently become a popular tool for DTI prediction and drug repositioning (Bleakley and Yamanishi, 2009; Chen *et al.*, 2012; Cheng *et al.*, 2012; Mei *et al.*, 2012; van Laarhoven *et al.*, 2011; Xia *et al.*, 2010; Xie *et al.*, 2012; Yamanishi *et al.*, 2008). Although these network-based approaches have achieved promising results, most of them can only predict binary DTIs, that is, they can only determine whether a drug interacts with a target protein, but cannot tell how they interact with each other. However, individual DTIs generally have different meanings. For example, drug-target pairs can be described by different *relationships*, such as *direct* and *indirect* interactions (Günther *et al.*, 2008). A direct interaction is usually caused by protein-ligand binding, whereas an indirect interaction can be induced by the changed expression

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level of a protein or active metabolites produced by a drug. In addition, DTIs can be annotated with different drug modes of action, e.g. activation and inhibition (Kuhn *et al.*, 2012). Here, ‘drug modes of action’ may be slightly different from the term used in the literature, e.g. (Iorio *et al.*, 2010). In this article, we mainly use this term to represent the following three specific types of interactions: binding, activation and inhibition. Hereinafter, we will use ‘types of DTIs’ to represent both drug-target relationships and drug modes of action. A network in which links are associated with different meanings is called the *multidimensional network* (Lü and Zhou, 2011). In our context, we call a DTI network where links are annotated with different types of interactions the *multidimensional DTI network*. On the one hand, types of DTIs provide additional useful information, which can be incorporated into a multidimensional network to improve DTI prediction. On the other hand, the predicted types of DTIs can extend our understanding about the molecular basis of drug action. Despite these positive aspects, *current rich information about types of DTIs* (Günther *et al.*, 2008; Kuhn *et al.*, 2012) *has not been well exploited for DTI prediction, and how to incorporate such information into a multidimensional network to predict different types of DTIs still remains an open question.*

In this article, we propose an effective machine-learning approach to accurately predict different types of DTIs on a multidimensional network. Unlike previous network-based approaches (Bleakley and Yamanishi, 2009; Chen *et al.*, 2012; Cheng *et al.*, 2012; Mei *et al.*, 2012; van Laarhoven *et al.*, 2011; Xia *et al.*, 2010; Yamanishi *et al.*, 2008), which only predict binary DTIs, our approach not only identifies new DTIs but also infers their corresponding types of interactions, such as drug-target relationships or drug modes of action. Our approach uses a generalized version of a two-layer undirected graphical model, called *restricted Boltzmann machine* (RBM) (Hinton and Salakhutdinov, 2006), to represent a multidimensional DTI network, which encodes different types of DTIs. In addition, we apply a practical learning algorithm, called *Contrastive Divergence* (CD) (Hinton, 2002), to train our RBM model and predict unknown types of DTIs.

To our knowledge, *our work is the first approach to predict different types of DTIs on a multidimensional network, which not only describes binary DTIs but also encodes their corresponding types of interactions.* We have tested our algorithm on two public databases, namely, MATADOR (Günther *et al.*, 2008) and STITCH (Kuhn *et al.*, 2012), which contain information about drug-target relationships and drug modes of action, respectively. Our tests demonstrate that our RBM model can be used as a highly powerful tool for integrating different types of DTIs into a multidimensional network and predicting different types of interactions with high accuracy. In particular, our results show that integrating different types of DTIs into prediction with distinction can achieve the area under precision-recall curve (AUPR) up to 89.6, which can significantly outperform other predictions either by simply mixing multiple types of interactions without distinction or using only a single interaction type. Further tests show that our approach can predict a high percentage of novel DTIs that has been validated by known experiments in the literature or other databases. These results indicate that our approach can have potential applications in drug repositioning and hence advance the drug discovery process.

1.1 Related work

A number of network-based approaches have been proposed for predicting unknown interactions between drugs and targets. In Yamanishi *et al.* (2008), a supervised learning framework was developed based on a bipartite graph, which integrates both chemical and genomic spaces by mapping them into a unified space. Cheng *et al.* (2012) proposed a network-based inference approach to predict new DTIs by exploiting the topology similarity of the underlying interaction network. In Zhao and Li (2010), drug phenotypic, chemical indexes and protein-protein interactions in genomic space were integrated into a computational framework for DTI prediction. Chen *et al.* (2012) proposed a random walk approach for DTI prediction based on a heterogeneous network, which integrates drug similarity, target similarity and DTI similarity. Bleakley and Yamanishi (2009) presented a new approach, called *Bipartite Local Model* (BLM), to predict unknown DTIs by combining independent drug-based and target-based prediction results using a supervised learning method. Mei *et al.* (2012) further extended this BLM approach to incorporate the capacity of learning from neighbors and predict the interactions for new drug or target candidates. In Xia *et al.* (2010), a manifold regularization semi-supervised learning method was proposed to integrate heterogeneous biological data sources for DTI prediction. A regularized least square algorithm was proposed in van Laarhoven *et al.* (2011) for DTI prediction using a product of kernels derived from DTI profiles. In Gottlieb *et al.* (2011), He *et al.* (2010) and Perlman *et al.* (2011), DTI prediction was formulated into a classification problem after defining multiple groups of drug-related and target-related features, such as drug-drug and gene-gene similarity measures. In addition to chemical and genomic data, phenotypic information, such as side-effect profiles (Campillos *et al.*, 2008; Mizutani *et al.*, 2012), transcriptional response data (Iorio *et al.*, 2010) and public gene expression data (Dudley *et al.*, 2011; Sirota *et al.*, 2011), has also been used for DTI prediction and drug repositioning. Although previous network-based approaches have achieved promising results for DTI prediction and drug repositioning, few of them are specifically designed for integrating and predicting different types of DTIs on a multidimensional network.

RBMs, which are used as important learning modules for constructing deep belief nets (Arel *et al.*, 2010; Bengio, 2009), have been successfully applied in many fields, such as dimensionality reduction (Hinton and Salakhutdinov, 2006), classification (Larochelle and Bengio, 2008), collaborative filtering (Salakhutdinov *et al.*, 2007) and computational biology (Eickholt and Cheng, 2012). Recently, the predictive power of RBMs has also been demonstrated in the Netflix Prize contest (Bell and Koren, 2007; Salakhutdinov *et al.*, 2007), a public competition for developing the best collaborative filtering algorithm to predict user ratings for movies. To our knowledge, our work is the first approach to apply RBMs into large-scale DTI prediction.

2 METHODS

2.1 RBMs for DTI prediction

We use an RBM to formulate the DTI prediction problem on a multidimensional network. An RBM is a two-layer graphical model that can be

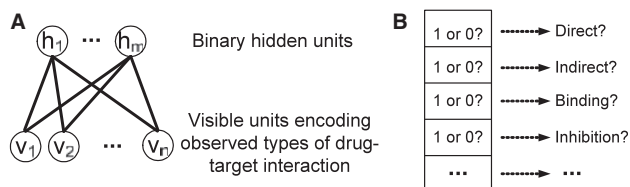


Fig. 1. An RBM with binary hidden units representing latent features and visible units encoding observed types of DTIs. (A) Overview of an RBM, where m is the number of hidden units and n is the number of visible units. (B) The information encoded in a visible unit

used to learn a probability distribution over input data (Hinton and Salakhutdinov, 2006). As shown in Figure 1A, an RBM consists of a layer of *visible* units and a layer of *hidden* units. Each visible unit is connected to all hidden units, and no intra-layer connection exists between any pair of visible units or any pair of hidden units. In our RBM model, visible units encode observed types of DTIs, such as drug-target relationships and drug modes of action (Fig. 1B), and hidden units represent latent features describing DTIs.

We use a simple example (Fig. 2) to describe how to construct RBMs from a multidimensional DTI network. Figure 2A shows a toy example of a multidimensional DTI network, where each DTI is associated with binary variables, which represent the states of direct and indirect interactions, respectively. As shown in Figure 2B, we build a specific RBM for every single target with the same number of hidden units and the same definitions of visible units. The binary states of visible variables for those missing DTIs are treated as zero in the constructed RBMs. The constructed RBMs for a multidimensional DTI network are associated with the same parameters. In other words, the constructed RBMs for all targets use the same parameters between hidden and visible layers. In Figure 2B, both RBMs for both target 1 and target 2 share the same parameters.

Compared with other prediction approaches, our RBM model can capture not only the correlations of drug-target pairs in a DTI network but also the correlations of different types of DTIs. Here, we use the simple example shown in Figure 2 to illustrate how our RBM model exploits the correlations in a multidimensional network to make prediction. Suppose that we have known different types of DTIs for target 1 in this example. To know which states of hidden units our RBM model should have, we let the three drugs send messages to hidden units and update their corresponding states. Conversely, once we know the states of hidden units for target 1, hidden units then send messages to visible units for the connected drugs and update their corresponding states. As in other collaborative filtering applications (Salakhutdinov *et al.*, 2007), after a number of iterations of the aforementioned updates, our RBM model can effectively capture the underlying features encoded in a multidimensional network. Based on the trained parameters between layers of hidden and visible units, our RBM model can then predict unknown types of DTIs based on input visible data.

In an RBM, suppose that in total there are n visible units, m hidden units and t types of DTI encoded in a visible unit. Let binary indicator vector $\mathbf{v}_i = (v_i^1, \dots, v_i^t)$, $i = 1, \dots, n$, denote state of the i -th visible unit, where *visible variables* $v_i^k = 1$ if the k -th type of DTI is observed in input data, and $v_i^k = 0$ otherwise. Let binary indicator variable h_j , $j = 1, \dots, m$, denote the state of the j -th hidden unit. Let W_{ij}^k be the weight associated with a connection between visible variable v_i^k and hidden variable h_j . Let $\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\mathbf{h} = (h_1, \dots, h_m)$ denote the configurations of visible and hidden layers, respectively. Then $(\mathbf{v}, \mathbf{h})$ is called a *joint configuration* of an RBM. As there is no intra-layer connection within both hidden and visible layers in an RBM, the energy of a

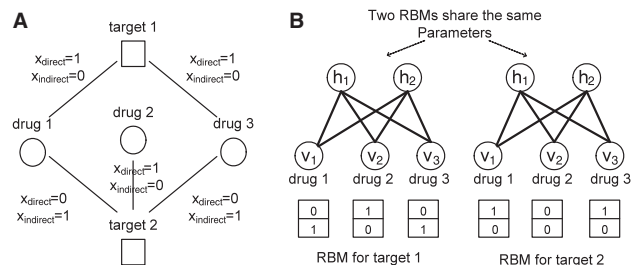


Fig. 2. A toy example for constructing RBMs from a multidimensional DTI network. (A) A simple multidimensional DTI network, where indicators x_{direct} and $x_{indirect}$ are equivalent to 1 if corresponding interaction types are present in visible data and 0 otherwise. (B) Constructed RBMs for corresponding targets. The binary numbers inside rectangles represent the states of visible variables. The RBMs for both target 1 and target 2 share the same parameters

joint configuration $(\mathbf{v}, \mathbf{h})$ can be defined by

$$E(\mathbf{v}, \mathbf{h}) = - \sum_{i=1}^n \sum_{k=1}^t a_i^k v_i^k - \sum_{j=1}^m b_j h_j - \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^t W_{ij}^k h_j v_i^k \quad (1)$$

where n , m and t are the numbers of visible units, hidden units and types of DTI encoded in a visible unit, respectively, v_i^k , h_j are visible variables and hidden units, respectively, and a_i^k , b_j are their corresponding bias terms (offsets), and W_{ij}^k is the corresponding weight for the connection between visible variable v_i^k and hidden variable h_j . Then, the probability of a joint configuration $(\mathbf{v}, \mathbf{h})$ can be defined by

$$\Pr(\mathbf{v}, \mathbf{h}) = \frac{1}{Z} \exp(-E(\mathbf{v}, \mathbf{h})) \quad (2)$$

where $Z = \sum_{\mathbf{v}, \mathbf{h}} \exp(-E(\mathbf{v}, \mathbf{h}))$ is called the *normalizing constant* or *partition function*. Summing over all possible configurations of $\mathbf{h}$, we obtain the following marginal distribution over visible data $\mathbf{v}$:

$$\Pr(\mathbf{v}) = \frac{1}{Z} \sum_{\mathbf{h}} \exp(-E(\mathbf{v}, \mathbf{h})). \quad (3)$$

As there is no intra-layer connection between any pair of visible or hidden units, we can define the following conditional probabilities:

$$\Pr(v_i^k = 1 | \mathbf{h}) = \sigma(a_i^k + \sum_{j=1}^m W_{ij}^k h_j) \quad (4)$$

$$\Pr(h_j = 1 | \mathbf{v}) = \sigma(b_j + \sum_{i=1}^n \sum_{k=1}^t W_{ij}^k v_i^k) \quad (5)$$

where n , m and t are the numbers of visible units, hidden units and types of DTI encoded in a visible unit, respectively, and $\sigma(x) = 1/(1 + e^{-x})$ is the logistic function. These conditional probabilities are important for the iterative updates between hidden and visible layers when training an RBM model.

Unlike the conditional probabilities of visible variables in Salakhutdinov *et al.* (2007), which used a conditional multinomial distribution, here, our conditional probability of visible variables in Equation (4) uses a conditional Bernoulli distribution. This is because in our RBM model, more than one visible variable can be assigned one for a DTI. For example, a DTI can be annotated with 'direct', 'binding' and 'inhibition' simultaneously. Our RBM model is data-driven and respect data as much as possible. As also demonstrated in Salakhutdinov *et al.* (2007), RBMs can capture the nature distribution of data. As we treat each interaction type as a separate binary variable, it is possible that two inconsistent interaction types (e.g. 'direct' and 'indirect') can be both yielded by our RBM model. This means that the data implies that these two interaction

types are both possible. In addition, by defining each interaction type as a separate variable, our RBM model is general and can be easily extended to include more interaction types from other data sources without much manual inspection on the consistence of different types of DTIs.

2.2 Training

We first use a real example to demonstrate the complexity of parameter training in our RBM model. In the MATADOR-based data that we have tested (Section 3.1), the number of drugs is $n = 784$. When only predicting direct and indirect interactions, the number of interaction types that we need to consider is $t = 2$. In our RBM model, we typically set the number of hidden units $m = 100$. In such a case, the total number of parameters in our RBM model is $n * t + m + n * m * t = 158468$. In general, training an RBM with such a large number of parameters poses a difficult task. Later in the text, we will describe how to perform parameter training in an RBM model.

To train an RBM and learn its parameter, we need to maximize the likelihood of visible data with respect to the parameters W_{ij}^k , a_i^k and b_j . To achieve this goal, we could perform gradient ascent in the log-likelihood of the training data derived from Equation (3) and incrementally adjust the weights and biases:

$$\Delta W_{ij}^k = \epsilon \frac{\partial \log \Pr(\mathbf{v})}{\partial W_{ij}^k} = \epsilon (\langle v_i^k h_j \rangle_{data} - \langle v_i^k h_j \rangle_{model}) \quad (6)$$

where ϵ is the *learning rate*, $\langle \cdot \rangle_{data}$ denotes an expectation of the data distribution and $\langle \cdot \rangle_{model}$ denotes an expectation of the distribution defined by the model.

In Equation (6), $\langle \cdot \rangle_{data}$ can be computed easily based on the frequency information obtained from visible data. Unfortunately, it is generally difficult to compute $\langle \cdot \rangle_{model}$, as it would require exponential time to do so. To avoid this problem, Hinton (2002) proposed a practical learning algorithm, called Contrastive Divergence (CD), which minimizes the Kullback–Leibler divergence. In this work, we use a mean-field version of the CD algorithm (Le Roux and Bengio, 2008; Welling and Hinton, 2002) to train our RBM model. In particular, we use the following procedure in each training pass to incrementally adjust the weights and biases:

$$\Delta W_{ij}^k = \epsilon (\langle v_i^k h_j \rangle_{data} - \langle v_i^k h_j \rangle_T) \quad (7)$$

$$\Delta a_i^k = \epsilon (\langle v_i^k \rangle_{data} - \langle v_i^k \rangle_T) \quad (8)$$

$$\Delta b_j = \epsilon (\langle h_j \rangle_{data} - \langle h_j \rangle_T) \quad (9)$$

where ϵ is the *learning rate*, $\langle \cdot \rangle_{data}$ denotes an average value over all input data for each update and $\langle \cdot \rangle_T$ denotes the average value over T mean-field iterations, which is considered a good approximation of $\langle \cdot \rangle_{model}$ in the log-likelihood function in Equation (6) (Hinton, 2002; Le Roux and Bengio, 2008; Welling and Hinton, 2002).

2.3 Making predictions

To predict the unknown types of DTIs for a given target with visible data $\mathbf{v}$, we compute the following probability distribution after one iteration of the mean-field updates (Salakhutdinov *et al.*, 2007):

$$\hat{p}_j = \Pr(h_j | \mathbf{v}) = \sigma(b_j + \sum_{i=1}^n \sum_{k=1}^t W_{ij}^k v_i^k) \quad (10)$$

$$\Pr(v_i^k = 1 | \hat{p}_1, \dots, \hat{p}_m) = \sigma(a_i^k + \sum_{j=1}^m W_{ij}^k \hat{p}_j), \quad (11)$$

where n , m and t are the numbers of visible units, hidden units and types of DTI encoded in a visible unit, respectively, W_{ij}^k , a_i^k and b_j are

parameters that have been learned in Section 2.2, and $\hat{p}_j$ is the conditional probability of hidden variable h_j given visible data. After that, we compute the expectation as our final prediction for the query target.

2.4 Conditional RBMs

For a drug-target pair, which does not have a connection on a multidimensional DTI network, it could be considered either a missing interaction or a negative interaction. In general, a set of known DTIs that have been verified experimentally provide more reliable information than those drug-target pairs, which do not have connections on a multidimensional DTI network. To incorporate this additional information, we use a *conditional RBM* (Salakhutdinov *et al.*, 2007) to further formulate our prediction problem. Let $\mathbf{r} = (r_1, \dots, r_m)$ be a binary indicator vector, in which $r_j = 1$ if there exists a known DTI between the input target and the j -th drug, and $r_j = 0$ otherwise. The states of r_j can be directly obtained based on the visible DTI data from a constructed DTI network. Here, $\mathbf{r}$ is a binary indicator vector representing the reliability of observed data. In general, DTIs that have been verified experimentally provide more reliable information than unknown DTIs. A conditional RBM defines a joint distribution over $(\mathbf{v}, \mathbf{h})$ conditioned on $\mathbf{r}$. In a conditional RBM, the conditional probabilities of visible and hidden units can be written as:

$$\Pr(h_j = 1 | \mathbf{v}, \mathbf{r}) = \sigma\left(b_j + \sum_{i=1}^n \sum_{k=1}^t W_{ij}^k v_i^k + \sum_{i=1}^m D_{ij} r_i\right), \quad (12)$$

where D_{ij} is a parameter describing the effect of $\mathbf{r}$ on $\mathbf{h}$. The parameter D_{ij} can be also learned using the CD algorithm:

$$\Delta D_{ij} = \epsilon (\langle h_j \rangle_{data} - \langle h_j \rangle_T) r_i. \quad (13)$$

2.5 Implementation

We chose a conditional RBM to perform DTI prediction on a multidimensional network that encodes different types of DTIs. The algorithm was implemented in Java, using the **jaRBM** package (<http://sourceforge.net/projects/jarbm/>). For the value of m , we chose an empirical range according to the literature. Here, we chose $m = 100$. We set learning rate $\epsilon = 0.01$. For other parameters, we chose the default values that were defined in the **jaRBM** package. The initial values of W_{ij}^k , b_j , a_i^k and D_{ij} were sampled from Gaussian distribution with standard derivation 0.1. In the CD learning procedure, we ran the mean-field updates 100 passes over training data. The training of our RBM model runs in several hours on a typical training dataset. For example, the training of our RBM model on the MATADOR-based data (Supplementary Table S1 in Section S1) takes 5.5 h on a PC machine with a 3.3 GHz Intel core i5 processor and 4 GB memory.

3 RESULTS AND DISCUSSION

3.1 Datasets and evaluation metrics

We tested our RBM model on two datasets, which were derived from MATADOR (Günther *et al.*, 2008) and STITCH (Kuhn *et al.*, 2012), respectively. MATADOR is a manually curated online database, which mainly describes drug-target relationships, including direct and indirect DTIs. The list of direct and indirect DTIs in MATADOR was first extracted by automated text-mining and then followed by manual annotation (Günther *et al.*, 2008). STITCH provides modes of action for the interactions between proteins and chemicals, which were annotated based on evidence derived from known experiments in the literature (Kuhn *et al.*, 2012). For the MATADOR-based dataset, we only extracted those DTI records in which protein and

drug names are present and annotation terms correspond to ‘DIRECT’ or ‘INDIRECT’. For the STITCH-based dataset, we only kept a list of DTI records in which drugs and target proteins overlap those in MATADOR. For the mode of action term in the STITCH-based data, we only considered ‘binding’, ‘activation’ and ‘inhibition’. In summary, the MATADOR-based dataset contains 784 drugs, 2431 protein targets and 13 064 DTIs, in which 7862 interactions are direct and 5202 interactions are indirect. The STITCH-based dataset contains 598 drugs, 671 protein targets and 3296 DTIs, in which 2589, 945 and 1493 interactions are annotated with ‘binding’, ‘activation’ and ‘inhibition’, respectively. Descriptive statistics about these two datasets can be found in Table S1 in Supplementary Material Section S1.

Receiver Operator Characteristic (ROC) and Precision-Recall (PR) curves are commonly used to assess the performance of a prediction model. For highly skewed data, ROC curves can give an overoptimistic picture of an algorithm’s performance (Davis and Goadrich, 2006). In this scenario, PR curves provide a better view of the prediction results than ROC curves. As stated in Bleakley and Yamanishi (2009) and van Laarhoven *et al.* (2011), when evaluating the DTI prediction results, in which there are usually few positive DTIs, PR curves provide greater biological significance and are considered a better quality measure than ROC curves. Therefore, we mainly used PR curves and the AUPR curve to evaluate the performance of our algorithm, though we also reported the area under the ROC curve (AUC) in our test results.

3.2 Predicting direct and indirect DTIs

In general, drug-target relationships can be classified into two categories, including *direct* and *indirect* interactions. Direct interactions are usually caused by protein-ligand binding, whereas indirect interactions can be driven by different mechanisms. For instance, indirect interactions can be induced by metabolites of drugs or changes in gene expressions (Günther *et al.*, 2008). To examine whether our algorithm can accurately predict direct and indirect DTIs on a multidimensional network, we tested it on the MATADOR-based data. In particular, we performed the following three tests: (i) Integrating both direct and indirect DTIs with distinction, which means that the input data of the RBM is a multidimensional vector indicating different types of interactions; (ii) Mixing both direct and indirect DTIs without distinction, which means that the input data is a one-dimensional binary vector indicating whether DTIs are observed; (iii) Using only a single interaction type (i.e. using only direct or indirect DTIs). For each test, we performed a 10-fold cross-validation procedure, as described in Supplementary Material Section S3.

The results of the aforementioned three tests are summarized in Table 1, and their corresponding PR curves are shown in Figure 3. When integrating both direct and indirect interactions with distinction, our algorithm achieved the best performance with an AUPR score up to 89.6 for direct interaction prediction and 80.1 for indirect interaction prediction. These results demonstrated that our RBM model can effectively integrate different drug-target relationships on a multidimensional DTI network to accurately predict direct or indirect interactions. In addition, our results showed that integrating both direct and indirect DTIs

Table 1. Results on predicting direct and indirect DTIs

Drug-target relationship	Test method	AUC	AUPR
Direct interaction	Integrating data with distinction	98.7	89.6
	Mixing data without distinction	98.8	72.1
	Using direct interactions only	98.0	78.9
Indirect interaction	Integrating data with distinction	97.1	80.1
	Mixing data without distinction	97.0	37.8
	Using indirect interactions only	94.8	62.4

Note: ‘Integrating data with distinction’ corresponds to the test in which our algorithm integrated both direct and indirect interactions with distinction. ‘Mixing data without distinction’ corresponds to the test in which our algorithm mixed both direct and indirect interactions without distinction. ‘Using direct (indirect) interactions only’ corresponds to the test in which our algorithm used only direct (indirect) interactions. The highest AUPR score is shown in bold.

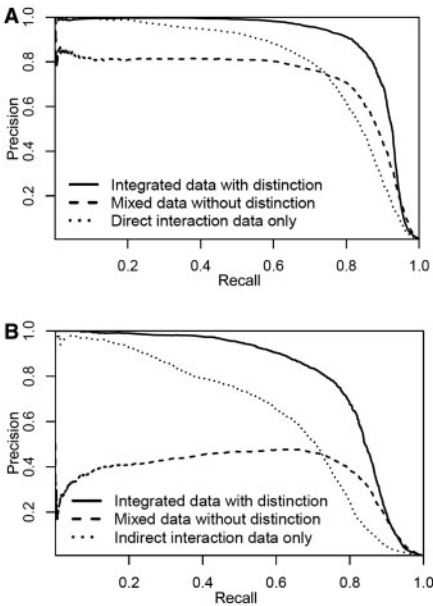


Fig. 3. PR curves for the direct and indirect DTIs predicted by our RBM model. (A) PR curves for the direct DTIs predicted by our model. (B) PR curves for the indirect DTIs predicted by our model

with distinction significantly outperformed the other two tests, which only considered a single interaction type or simply mixed direct and indirect interactions without distinction. In particular, the integration of direct and indirect interactions improved the AUPR score by >10% for direct interaction prediction and >17% for indirect interaction prediction (Table 1). In PR curves, compared with the other two tests, integrating both interaction types with distinction achieved a better precision value almost at every recall value (Fig. 3). An interesting result is that mixing both interactions without distinction yielded worse performance than using a single interaction type (Table 1). For example, for direct interaction prediction, the AUPR score was decreased from 78.9 to 72.1. This result implies that incorporating a new data type into DTI prediction is non-trivial and requires

careful data integration. In addition, our results show that our RBM model respect the data and can predict a high percentage of consistent DTIs. For example, when we chose 0.5 as the probability threshold to infer the interaction type, >99% of positive DTIs predicted by our algorithm were consistent (i.e. <1% of DTIs were predicted as both direct and indirect interactions).

In our 10-fold cross-validation test (Table 1), the test method ‘using direct (indirect) interaction only’ used less training data than other two test methods. This may create bias when comparing two prediction methods using training data with different sizes. To make a fair comparison, we have made an additional comparison between methods ‘integrating data with distinction’ and ‘using direct (indirect) interaction only’ using training data of the same size (Supplementary Material Section S4). Our new comparison results confirmed that integrating data with distinction outperformed the method that uses a single interaction only.

To check whether our algorithm can have a wider range of applications, we also performed a 5-fold cross-validation test (Supplementary Material Section S3). Compared with the 10-fold cross-validation results, we only found a small decrease in our algorithm’s performance in the 5-fold cross-validation test.

To perform more sanity check on our algorithm’s performance, we also conducted another test that is similar to leave-one-out cross-validation except that we removed homologous proteins from training data. For training data, we only kept those DTIs in which proteins have sequence identity <25% to the target of the validating interaction. This process can significantly reduce the size of dataset. In our test, the training dataset has only ~300 targets. This test is more rigorous than previous leave-one-out cross-validation (LOOCV) tests performed in (Bleakley and Yamanishi, 2009; van Laarhoven *et al.*, 2011; Yamanishi *et al.*, 2008), in that we have removed similar proteins with high sequence identity from training data and thus reduced impact of protein homology on the network-based prediction of DTIs.

The results of our LOOCV like test show that, our approach still achieved decent prediction accuracy, with AUPR 79.0 for direct interaction prediction, but the AUPR score for indirect interaction prediction dropped to 59.1. We further investigated this problem and found that the average degree of indirect interactions with drugs for targets in training data was reduced to 1.67. With insufficient number of DTIs, it would be difficult to train our RBM model and make accurate prediction.

To mimic the real situation in which DTI network data are available, we also performed an additional test on those DTIs in which the degree of the target is above the average degree of the multidimensional network (i.e. the degree of the target in the validating interaction is ≥ 6). The results of this test show that, our approach achieved decent performance, with AUPR 80.4 for direct interaction prediction and 74.5 for indirect interaction prediction. These results indicate that in the real situation, our RBM model can make reasonably accurate prediction even homologous proteins are not present in training data.

As little work has been developed for predicting unknown types of DTIs in a multidimensional network, it is difficult for us to directly compare our work with other prediction approaches. Instead, we have compared our algorithm with a simple logic-based approach on the MATADOR-based data. This simple logic-based approach follows the basic premise (i.e. similar drugs and targets should have similar interactions) that

has been widely used in previous prediction approaches. More details of this simple logic-based algorithm can be found in Supplementary Material Section S5. As summarized in Table 2, the comparison results show that our algorithm outperformed the simple logic based approach. This is expected, as the simple logic based approach simply uses the closest interaction type profiles to predict unknown types of DTIs. Such a strategy only exploits a small proportion of training data and cannot be sufficient enough to capture the deep correlations of different types of DTIs in a multidimensional network. On the other hand, our RBM model uses hidden units to represent the intrinsic correlations of different types of DTIs in the network and can effectively capture the latent feature of drug-target relationships and thus make accurate predictions.

Overall, our approach achieved better performance for direct interaction prediction than for indirect interaction prediction (Table 1). Probably this is because in the MATADOR-based data, indirect interactions describe many different mechanisms of DTIs and thus provide less predictive power than direct interactions. We expect that our method would achieve better performance for indirect interaction prediction if different mechanisms of indirect DTIs in the MATADOR-based data could be further identified.

3.3 Predicting drug modes of action

In addition to direct and indirect interactions, DTIs can be also annotated with different drug modes of action, such as activation or inhibition. To evaluate our algorithm’s performance on predicting different drug modes of action, we tested it mainly on the STITCH-based data. We focused on three drug modes of action, including binding, activation and inhibition. In addition to testing the STITCH-based data, we included two additional tests, which further incorporated the MATADOR-based data, that is, each DTI in the STITCH-based data was also associated with direct or indirect interaction derived from the MATADOR-based data. Overall, we conducted the following five tests: (i) integrating both drug-target relationships from the MATADOR-based data and drug modes of action from the STITCH-based data with distinction; (ii) mixing DTIs from both MATADOR-based and STITCH-based data without distinction (i.e. all DTIs were associated with only binary values); (iii) integrating drug modes of action from the STITCH-based data with distinction; (iv) mixing drug modes of action from the STITCH-based data without distinction; and (v) using a single mode of action from the STITCH-based data. For every test, we

Table 2. Results on comparing our approach with the simple logic based approach

Drug-target relationship	Test method	AUC	AUPR
Direct interaction	Our approach	98.7	89.6
	Simple logic-based approach	92.1	81.6
Indirect interaction	Our approach	97.1	80.1
	Simple logic-based approach	88.4	74.5

Note: The highest AUPR score is shown in bold.

also carried out a 10-fold cross-validation procedure as described in Supplementary Material Section S3.

Both AUC and AUPR scores of the aforementioned five tests are reported in Table 3, and their corresponding PR curves are shown in Figure 4. When integrating different types of DTIs with distinction, our algorithm can achieve AUC up to 96.9 and AUPR up to 79.1. The test results showed that our RBM model can effectively exploit different types of interactions encoded in a multidimensional DTI network to predict drug modes of action. Among all five tests, integrating different types of DTIs with distinction produced higher AUPR scores than the other tests, which used only a single interaction type or simply mixed types of interactions without distinction. When differentiating different types of DTIs, integrating direct and indirect interactions from the MATADOR-based data slightly improved the prediction performance. When comparing the results on predicting different modes of action, binding interactions were predicted with higher accuracy than activation and inhibition interactions. This is probably because in the STITCH-based data, binding interactions are more informative in predicting DTIs than other two modes of action.

As in Section 3.2, we also performed an additional test using training data of the same size for comparing methods ‘integrating data with distinction’ and ‘using a single data type only’, when predicting different modes of action. Our new comparison results confirmed that integrating different data types with distinction outperformed the method that uses only a single data type (Supplementary Material Section S4).

3.4 New predictions

To examine our algorithm’s ability for predicting novel DTIs that are not present in our data derived from MATADOR, we

performed the following test, which is similar to the experiments conducted in (Bleakley and Yamanishi, 2009; van Laarhoven *et al.*, 2011). We predicted unknown direct DTIs using the complete MATADOR-based dataset as training data. For the predicted results, we mainly focused on DTIs involving those proteins that are not listed as drug-metabolizing enzymes or transporter proteins in DrugBank (Knox *et al.*, 2011). We outputted the set of the top 50 scoring predictions that are not present in training data. We then checked whether these new predicted DTIs appear in other databases, including ChEMBL (Gaulton *et al.*, 2011), DrugBank (Knox *et al.*, 2011) and STITCH (Kuhn *et al.*, 2012).

Supplementary Figure S1 in Section S2 visualizes part of the DTI network constructed based on the set of the 50 highest scoring interactions predicted by our approach using the MATADOR-based data. Table 4 shows the list of the top 10 scoring direct DTI predictions. Among these predicted DTIs, a high fraction of the predicted interactions (7 of 10) was found in ChEMBL, DrugBank or STITCH. In the remaining three DTIs that we did not find experimental evidence from the three databases, there might still exist interaction for each drug-target pair. For example, although it is known that mifepristone does not bind to estrogen receptor, it may have some effects on the expression of estrogen receptor (Jiang *et al.*, 2007).

In Table 4, some of our new predictions are trivial, as they can also be easily derived based on known interactions of similar proteins or drugs in the dataset. However, our new predictions also provided some interesting results. For example, among the top 50 scoring predictions, our algorithm suggested that there may exist interactions between spironolactone and the membrane progesterin receptor gamma protein and between mesalazine (also called 5-aminosalicylic acid) and leukotriene-A4 hydrolase

Table 3. Results on predicting drug modes of action

Mode of action	Test method	AUC	AUPR
Binding interaction	Integrating MATADOR and STITCH with distinction	96.9	79.1
	Mixing MATADOR and STITCH without distinction	97.9	53.3
	Integrating data with distinction	95.0	78.1
	Mixing data without distinction	95.6	68.0
	Using binding interactions only	94.1	74.4
Activation interaction	Integrating MATADOR and STITCH with distinction	94.4	67.4
	Mixing MATADOR and STITCH without distinction	96.9	35.6
	Integrating data with distinction	91.2	65.2
	Mixing data without distinction	94.2	50.5
	Using activation interactions only	87.7	56.3
Inhibition interaction	Integrating MATADOR and STITCH with distinction	94.1	67.1
	Mixing MATADOR and STITCH without distinction	96.9	38.6
	Integrating data with distinction	92.5	65.2
	Mixing data without distinction	93.9	44.3
	Using inhibition interactions only	89.5	60.2

Note: ‘Integrating MATADOR and STITCH with distinction’ corresponds to the test in which our algorithm integrated both drug-target relationships from the MATADOR-based data and drug modes of action from the STITCH-based data with distinction. ‘Mixing MATADOR and STITCH without distinction’ corresponds to the test in which our algorithm mixed DTIs from both MATADOR-based and STITCH-based data without distinction. ‘Integrating data with distinction’ corresponds to the test in which our algorithm integrated drug modes of action from the STITCH-based data with distinction. ‘Mixing data without distinction’ corresponds to the test in which our algorithm mixed drug modes of action from the STITCH-based data without distinction. ‘Using binding (activation or inhibition) interactions only’ corresponds to the test in which our algorithm used only binding (activation or inhibition) interactions from the STITCH-based data. The highest AUPR score is shown in bold.

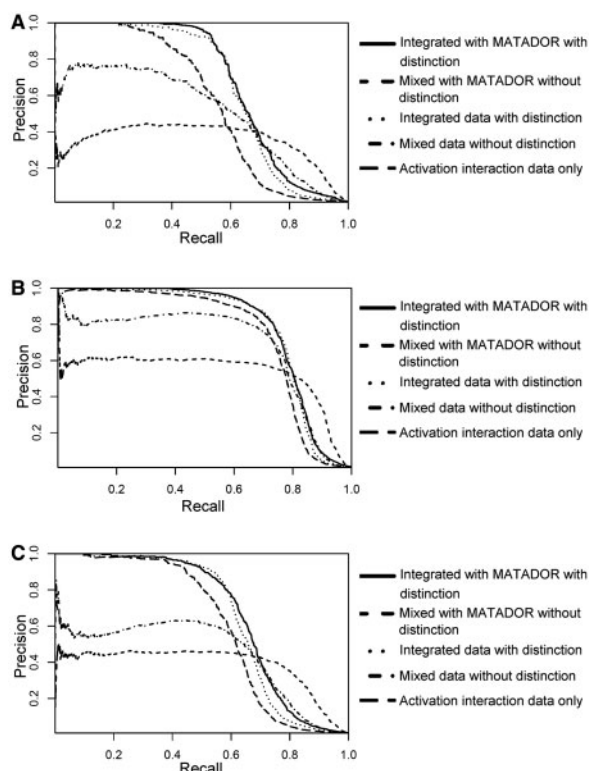


Fig. 4. PR curves for the predicted drug modes of action. (A) PR curves for the predicted binding interactions. (B) PR curves for the predicted activation interactions. (C) PR curves for the predicted inhibition interactions

(LTA4H) enzyme. Spironolactone is known as a diuretic or anti-hypertensive drug and can act on the aldosterone receptor as a competitive antagonist (Macdonald, 1997). Our predicted interaction between spironolactone and membrane progesterin receptor gamma protein indicates that spironolactone may have some progestogenic function. Interestingly, this hypothesis can be confirmed from the clinical studies performed in (Nakhjavani *et al.*, 2009). In addition, mesalazine is an anti-inflammatory drug that is primarily used to treat inflammatory bowel disease (Sandborn *et al.*, 2007). LTA4H is an important enzyme that converts leukotriene-A4 to leukotriene-B4 (Rudberg *et al.*, 2004). It has been proposed that leukotriene-B4 may play an important role in a number of different acute and chronic inflammatory diseases, including inflammatory bowel disease (Haeggström, 2000). These studies imply that our predicted interaction between mesalazine and LTA4H is probably true.

These results on new predictions indicated that our RBM model is practically useful in predicting novel DTIs and can have potential applications in drug repositioning.

4 CONCLUSION

In this article, we proposed a first machine-learning approach to predict different types of DTIs on a multidimensional network. Our approach uses an RBM model to effectively encode multiple sources of information about DTIs and accurately predict

Table 4. Top 10 scoring direct DTIs predicted by our approach using the MATADOR data

Rank	Pair ^a	Description	Evidence ^b
1	DB00812 P23219	Phenylbutazone PTGS1: Prostaglandin G/H synthase 1	C, D, S
2	DB04599 P42261	Aniracetam GRIA1: Glutamate receptor 1 precursor	S
3	DB00834 P03372	Mifepristone ESR1: Estrogen receptor	
4	DB01392 P28222	Yohimbine HTR1B: 5-hydroxytryptamine 1B receptor	D, S
5	DB01297 P07550	Practolol ADRB2: Beta-2 adrenergic receptor	S
6	DB01297 P13945	Practolol ADRB3: Beta-3 adrenergic receptor	
7	DB00334 P08908	Olanzapine HTR1A: 5-hydroxytryptamine 1 A receptor	D, S
8	DB01224 P21918	Quetiapine DRD5: D(1B) dopamine receptor	D
9	DB01224 P21728	Quetiapine DRD1: D(1 A) dopamine receptor	D
10	DB01233 P21918	Metoclopramide DRD5: D(1B) dopamine receptor	

^aDrugs and targets are represented by DrugBank IDs and UniProt ID, respectively.

^bDTIs that are observed in ChEMBL, DrugBank and STITCH are marked with 'C', 'D' and 'S', respectively.

different types of DTIs, such as drug-target relationships or drug modes of action. Tests on two public databases showed that our algorithm can achieve excellent prediction performance with high AUPR scores. Further tests indicated that our approach can infer a list of novel DTIs, which is practically useful for drug repositioning.

Although our algorithm has been tested only on direct and indirect drug-target relationships, and three drug modes of action, it is general and can be easily extended to integrate other types of DTIs (e.g. phenotypic effects). Current version of our prediction algorithm only considers connections between drugs and targets. In the future, we will extend our approach to exploit the connections within target proteins or drugs. For example, the sequence similarity scores between target proteins, the substructure similarity scores between drugs or drug-drug interactions (Gottlieb *et al.*, 2012; Tatonetti *et al.*, 2012) can be also incorporated into our prediction model. As the conventional version of an RBM does not allow the connections within the same layer, such an extension will require careful thought. Currently, our algorithm has been tested only on two databases (i.e. MATADOR and STITCH). We will test it on more data in the future. For example, it will have more significance if we can predict DTIs based on human proteins and all molecules in PubChem (Kaiser, 2005) or a similar database. Finally, we are also seeking wet-laboratory collaborators to experimentally verify the highest scoring DTIs predicted by our algorithm.

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REFERENCES

- Arel, I. *et al.* (2010) Deep machine learning—a new frontier in artificial intelligence research. *IEEE Comput. Intell. Mag.*, **5**, 13–18.
- Bell, R. and Koren, Y. (2007) Lessons from the netflix prize challenge. *ACM SIGKDD Explor. Newsl.*, **9**, 75–79.
- Bengio, Y. (2009) Learning deep architectures for AI. *Found. Trends Mach. Learn.*, **2**, 1–127.
- Bleakley, K. and Yamanishi, Y. (2009) Supervised prediction of drug-target interactions using bipartite local models. *Bioinformatics*, **25**, 2397–2403.
- Booth, B. and Zimmel, R. (2004) Prospects for productivity. *Nat. Rev. Drug. Discov.*, **3**, 451–456.
- Campillos, M. *et al.* (2008) Drug target identification using side-effect similarity. *Science*, **321**, 263–266.
- Chen, X. *et al.* (2012) Drug-target interaction prediction by random walk on the heterogeneous network. *Mol. Biosyst.*, **8**, 1970–1978.
- Cheng, A.C. *et al.* (2007) Structure-based maximal affinity model predicts small-molecule druggability. *Nat. Biotechnol.*, **25**, 71–75.
- Cheng, F. *et al.* (2012) Prediction of drug-target interactions and drug repositioning via network-based inference. *PLoS Comput. Biol.*, **8**, e1002503.
- Davis, J. and Goadrich, M. (2006) The relationship between precision-recall and roc curves. In: *Proceedings of the 23rd international conference on Machine learning*, ICML'06. ACM, New York, NY, USA, pp. 233–240.
- Dimasi, J.A. (2001) New drug development in the united states from 1963 to 1999. *Clin. Pharmacol. Ther.*, **69**, 286–296.
- Donald, B.R. (2011) *Algorithms in Structural Molecular Biology*. MIT Press, Cambridge, MA, USA.
- Dudley, J.T. *et al.* (2011) Computational repositioning of the anticonvulsant topiramate for inflammatory bowel disease. *Sci. Transl. Med.*, **3**, 96ra76.
- Eickholt, J. and Cheng, J. (2012) Predicting protein residue-residue contacts using deep networks and boosting. *Bioinformatics*, **28**, 3066–3072.
- Gaulton, A. *et al.* (2011) ChEMBL: a large-scale bioactivity database for drug discovery. *Nucleic Acids Res.*, **40**, D1100–D1107.
- Gottlieb, A. *et al.* (2011) PREDICT: a method for inferring novel drug indications with application to personalized medicine. *Mol. Syst. Biol.*, **7**, 496.
- Gottlieb, A. *et al.* (2012) INDI: a computational framework for inferring drug interactions and their associated recommendations. *Mol. Syst. Biol.*, **8**, 592.
- Günther, S. *et al.* (2008) SuperTarget and Matador: resources for exploring drug-target relationships. *Nucleic Acids Res.*, **36**, D919–D922.
- Haeggström, J.Z. (2000) Structure, function, and regulation of leukotriene A4 hydrolase. *Am. J. Respir. Crit. Care. Med.*, **161**, S25–S31.
- He, Z. *et al.* (2010) Predicting drug-target interaction networks based on functional groups and biological features. *PLoS One*, **5**, e9603.
- Hinton, G. and Salakhutdinov, R. (2006) Reducing the dimensionality of data with neural networks. *Science*, **28**, 504–507.
- Hinton, G.E. (2002) Training products of experts by minimizing contrastive divergence. *Neural Comput.*, **14**, 1771–1800.
- Iorio, F. *et al.* (2010) Discovery of drug mode of action and drug repositioning from transcriptional responses. *Proc. Natl Acad. Sci. USA*, **107**, 14621–14626.
- Jiang, J. *et al.* (2007) Effects of mifepristone on expression of estrogen receptor and progesterone receptor in cultured human eutopic and ectopic endometria. *Zhonghua Fu Chan Ke Za Zhi*, **36**, 218–221.
- Kaiser, J. (2005) Science resources: chemists want NIH to curtail database. *Science*, **308**, 774.
- Keiser, M.J. *et al.* (2007) Relating protein pharmacology by ligand chemistry. *Nat. Biotechnol.*, **25**, 197–206.
- Keiser, M.J. *et al.* (2009) Predicting new molecular targets for known drugs. *Nature*, **462**, 175–181.
- Knox, C. *et al.* (2011) DrugBank 3.0: a comprehensive resource for 'omics' research on drugs. *Nucleic Acids Res.*, **39**, D1035–D1041.
- Kroeze, W.K. *et al.* (2002) Molecular biology of serotonin receptors structure and function at the molecular level. *Curr. Top. Med. Chem.*, **2**, 507–528.
- Kuhn, M. *et al.* (2012) STITCH 3: zooming in on protein-chemical interactions. *Nucleic Acids Res.*, **40**, D876–D880.
- Larochelle, H. and Bengio, Y. (2008) Classification using discriminative restricted boltzmann machines. In: *Proceedings of the 25th international conference on Machine learning*, ICML'08. ACM, New York, NY, USA, pp. 536–543.
- Le Roux, N. and Bengio, Y. (2008) Representational power of restricted boltzmann machines and deep belief networks. *Neural Comput.*, **20**, 1631–1649.
- Lü, L. and Zhou, T. (2011) Link prediction in complex networks: a survey. *Physica A Stat. Mech. Appl.*, **390**, 1150–1170.
- Macdonald, F. (1997) *Dictionary of Pharmacological Agents*. CRC Press, Taylor and Francis Group, LLC, Boca Raton, FL, USA.
- MacDonald, M.L. *et al.* (2006) Identifying off-target effects and hidden phenotypes of drugs in human cells. *Nat. Chem. Biol.*, **2**, 329–337.
- Mei, J.P. *et al.* (2012) Drug-target interaction prediction by learning from local information and neighbors. *Bioinformatics*, **29**, 238–245.
- Mizutani, S. *et al.* (2012) Relating drug-protein interaction network with drug side effects. *Bioinformatics*, **28**, i522–i528.
- Morris, G.M. *et al.* (2009) AutoDock4 and AutoDockTools4: automated docking with selective receptor flexibility. *J. Comput. Chem.*, **30**, 2785–2791.
- Nakhjavani, M. *et al.* (2009) Short term effects of spironolactone on blood lipid profile: a 3-month study on a cohort of young women with hirsutism. *Br. J. Clin. Pharmacol.*, **68**, 634–637.
- Perlman, L. *et al.* (2011) Combining drug and gene similarity measures for drug-target elucidation. *J. Comput. Biol.*, **18**, 133–145.
- Roth, B.L. *et al.* (2004) Magic shotguns versus magic bullets: selectively non-selective drugs for mood disorders and schizophrenia. *Nat. Rev. Drug Discov.*, **3**, 353–359.
- Rudberg, P.C. *et al.* (2004) Leukotriene A4 hydrolase: identification of a common carboxylate recognition site for the epoxide hydrolase and aminopeptidase substrates. *J. Biol. Chem.*, **279**, 27376–27382.
- Salakhutdinov, R. *et al.* (2007) Restricted boltzmann machines for collaborative filtering. In: *Proceedings of the 24th international conference on Machine learning*, ICML'07. ACM, New York, NY, USA, pp. 791–798.
- Sandborn, W.J. *et al.* (2007) Medical management of mild to moderate crohn's disease: evidence-based treatment algorithms for induction and maintenance of remission. *Aliment Pharmacol. Ther.*, **26**, 987–1003.
- Sirota, M. *et al.* (2011) Discovery and preclinical validation of drug indications using compendia of public gene expression data. *Sci. Transl. Med.*, **3**, 96ra77.
- Tatonetti, N.P. *et al.* (2012) Data-driven prediction of drug effects and interactions. *Sci. Transl. Med.*, **4**, 125–131.
- Van Laarhoven, T. *et al.* (2011) Gaussian interaction profile kernels for predicting drug-target interaction. *Bioinformatics*, **27**, 3036–3043.
- Welling, M. and Hinton, G. (2002) A new learning algorithm for mean field boltzmann machines. *Artif. Neural Netw.*, **2415**, 351–357.
- Xia, Z. *et al.* (2010) Semi-supervised drug-protein interaction prediction from heterogeneous biological spaces. *BMC Syst. Biol.*, **4** (Suppl. 2), S6.
- Xie, L. *et al.* (2012) Novel computational approaches to polypharmacology as a means to define responses to individual drugs. *Annu. Rev. Pharmacol. Toxicol.*, **52**, 361–379.
- Yabuuchi, H. *et al.* (2011) Analysis of multiple compound-protein interactions reveals novel bioactive molecules. *Mol. Syst. Biol.*, **7**, 472.
- Yamanishi, Y. *et al.* (2008) Prediction of drug-target interaction networks from the integration of chemical and genomic spaces. *Bioinformatics*, **24**, i232–i240.
- Zhao, S. and Li, S. (2010) Network-based relating pharmacological and genomic spaces for drug target identification. *PLoS One*, **5**, e11764.